Encryption in Azure Databricks: at rest, in transit, and who holds the key

Encryption keeps data unreadable to anyone who gets around access controls. In Azure Databricks it is on by default everywhere; **customer-managed keys** change who controls the key.

> LEARNING OBJECTIVE
>
> Describe encryption at rest and in transit across the control plane, compute and storage, and the scope, rotation and operational obligations of customer-managed keys.

01 Why: what goes wrong without encryption

Why do we need encryption?

Permissions control who can open data. Encryption protects it when someone gets it another way, like a copied disk or an intercepted connection. Without the key, the data can't be read.

RISK	PROTECTION
A disk, snapshot or backup is copied or leaves the data centre	**At rest.** Stored bytes are AES-256 ciphertext; the copy is useless without the key.
Traffic between browser, cluster and storage is intercepted	**In transit.** TLS encrypts every hop, so captured packets reveal nothing.
A regulator or customer requires you to control who can decrypt	**Customer-managed key.** The key lives in your vault, under your policies and audit logs.
A contract ends or a breach needs containing now	**Revoke the key.** Disabling your key makes the data unreadable everywhere at once.

THREE QUESTIONS FOR ANY DATA

Where does it rest?

Which path does it travel?

Who holds the key?

02 Where data lives, and how it is encrypted

An Azure Databricks deployment has three places data rests: the **control plane** run by Databricks, the **compute plane** where clusters run, and **storage** in your subscription. Everything is encrypted at rest by default. The only question is whose key is used.

WHAT IS A CUSTOMER-MANAGED KEY?

By default, Microsoft and Databricks create and hold the keys that encrypt your data. With a **customer-managed key (CMK)**, you create the key in **your own Azure Key Vault** and let the service use it. The data is still encrypted the same way (AES-256). What changes is who owns the key and who can take it away.

1 · CREATE

You create an RSA key in Key Vault, e.g. `kv-data-prod/cmk-dbw`.

2 · GRANT

You give the service's identity permission to wrap and unwrap with the key. It can never export it.

3 · USE

The service encrypts data with its own data key, then locks that data key with your key.

4 · CONTROL

You rotate the key on your schedule. Disable it and the data can no longer be read.

Each store in the three zones below shows its default key and, in blue, where a CMK can be applied. Section 03 covers each scope in detail.

CONTROL PLANE · DATABRICKS SUBSCRIPTION

Control plane

Web app, APIs, job scheduler, notebooks and queries. Operated by Databricks.

Workspace metadata

Notebook source, interactive results, SQL queries and history, Databricks-backed secrets, Git credentials.

DEFAULT

Databricks-managed key

Encrypted by default; Databricks holds the key.

WITH CMK

CMK · managed services

Your Key Vault key wraps the encryption keys.

COMPUTE PLANE · CLUSTERS

Compute plane

Classic compute runs in a VNet in your subscription. Serverless runs in Databricks' subscription.

Classic cluster disks

VM OS and data disks: Delta cache and Spark shuffle spill.

DEFAULT

Platform-managed key

Azure disk encryption at rest, AES-256.

WITH CMK

CMK · managed disks

Disk encryption set uses your key.

Serverless compute

Short-lived VMs in Databricks' subscription, wiped after use.

DEFAULT

Databricks-managed key

Encrypted at rest and isolated per workload.

WITH CMK

Databricks-managed key

Managed-disk CMK does not apply to serverless.

STORAGE · YOUR SUBSCRIPTION

Storage

Where table data and files actually rest.

Workspace storage account

In the managed resource group: DBFS root, cluster logs, job results.

DEFAULT

Microsoft-managed key

Azure Storage encryption, AES-256.

WITH CMK

CMK · workspace storage

Set through the workspace encryption setting.

ADLS Gen2 (Unity Catalog)

Managed and external tables and volumes: stdatalake.

DEFAULT

Microsoft-managed key

Azure Storage encryption, AES-256.

WITH CMK

CMK on the storage account

Configured in Azure Storage, not Databricks.

IN TRANSIT · EVERY HOP

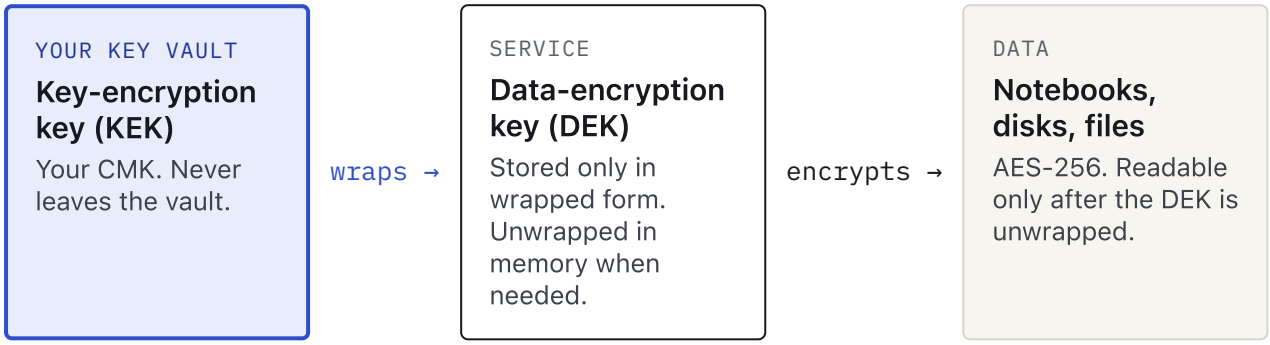
Data moving between zones travels over TLS. Private Link keeps the same traffic off the public internet, which is network isolation, not extra encryption.

1	**User or API client → control plane** `browser, CLI, BI tool`	`TLS 1.2+`	HTTPS for the workspace UI and REST APIs. Add front-end Private Link to keep it on private networks.
2	**Control plane ↔ classic compute** `secure cluster connectivity`	`TLS 1.2+`	Clusters open an outbound TLS tunnel to the control plane relay. No inbound ports and no public IPs on nodes.
3	**Compute → storage and services** `abfss://, Key Vault, JDBC`	`TLS 1.2+`	abfss:// is HTTPS-only. Set the storage account to require secure transfer and minimum TLS 1.2; set encrypt=true on JDBC sources.
4	**Worker ↔ worker** `Spark shuffle inside the VNet`	`Not by default`	Traffic stays inside your VNet. If policy requires it, enable inter-node encryption with an init script and accept a performance cost.

03 Customer-managed keys

With a customer-managed key (CMK), the key that protects your data lives in **your Azure Key Vault** or Managed HSM. You decide who can use it, when it rotates, and you can revoke it. That control comes with the duty to keep it available: lose the key and the data is gone.

HOW IT WORKS · ENVELOPE ENCRYPTION



Because only the small DEK is wrapped by your key, rotating the CMK re-wraps the DEK. Terabytes of data don't have to be re-encrypted.

Azure Databricks offers CMK for three scopes (Premium plan). Data in your own ADLS account is covered by Azure Storage's own CMK setting.

Managed services `Control plane`

Protects the data Databricks keeps for you in its control plane: your code, queries and stored credentials.

ENCRYPTS

- Notebook source and interactive notebook results stored in the control plane
- Secrets in Databricks-backed secret scopes
- Databricks SQL queries and query history
- Personal access tokens and Git credentials for Git folders

WHO NEEDS KEY ACCESS

`AzureDatabricks (first-party enterprise app)`

Get, Wrap Key and Unwrap Key on the key. With Key Vault RBAC: Key Vault Crypto Service Encryption User.

ROTATION

Create a new key version in Key Vault, then update the workspace to reference it. Keep earlier versions enabled until the update completes.

IF THE KEY IS DISABLED OR LOST

Notebooks, queries and secrets become unreadable and workspace features that use them stop working. Restoring the key restores access.

`Set on: Workspace encryption setting (portal, ARM, Bicep or Terraform)`

Workspace storage

Storage · managed resource group

Protects the workspace storage account Databricks creates in the managed resource group: DBFS root and workspace system data.

ENCRYPTS

- DBFS root (legacy file storage)
- Workspace system data such as job results and cluster logs
- Default workspace catalog storage, where used

WHO NEEDS KEY ACCESS

`Managed identity of the workspace storage account`

Get, Wrap Key and Unwrap Key on the key. Created when CMK is enabled; grant it access before applying the key.

ROTATION

Reference the key without a version to follow new versions automatically, or update to a specific version on your schedule.

IF THE KEY IS DISABLED OR LOST

Clusters can no longer read or write workspace storage; jobs and DBFS paths fail. Data in your own ADLS accounts is unaffected.

Set on: Workspace encryption setting

Managed disks `Compute plane · classic`

Protects the OS and data disks attached to classic cluster VMs, where Spark caches data and spills shuffle files.

ENCRYPTS

> Cluster VM managed disks in the managed resource group

> Delta cache and Spark shuffle spill written to local disk

WHO NEEDS KEY ACCESS

`Disk encryption set created for the workspace`

Get, Wrap Key and Unwrap Key on the key.

ROTATION

Supports automatic rotation: enable it and new clusters use the latest key version. Running clusters keep the version they started with.

IF THE KEY IS DISABLED OR LOST

New clusters fail to start. Serverless compute is not covered by this setting.

`Set on: Workspace encryption setting`

Your ADLS Gen2 Storage · your subscription

Unity Catalog managed and external tables live in storage accounts you own. Their encryption is configured on the storage account, outside Databricks.

ENCRYPTS

> Delta tables and volumes in Unity Catalog storage

> Any external location reached through a storage credential

WHO NEEDS KEY ACCESS

`User-assigned managed identity chosen on the storage account`

Key Vault Crypto Service Encryption User on the vault or key.

ROTATION

Enable automatic key version updates on the storage account, or set a Key Vault rotation policy.

IF THE KEY IS DISABLED OR LOST

All reads and writes to the account fail, from Databricks and every other service.

Set on: Storage account → Encryption (Azure Storage, not Databricks)

OPERATIONAL OBLIGATIONS · WHAT YOU SIGN UP FOR

1. **Protect the vault**
   Soft delete and purge protection are required. They are the only way back from an accidental key deletion.
2. **Never delete a key in use**
   Deleting or purging the key is crypto-shredding: the data cannot be recovered by you, Microsoft or Databricks.
3. **Rotate on a schedule**
   Set a Key Vault rotation policy and expiry alerts. Keep old versions enabled until every scope uses the new one.
4. **Keep it reachable**
   If the vault firewall blocks the service or the vault is unavailable, the workspace fails. Allow trusted Microsoft services or private endpoints.
5. **Separate duties**
   Key owners (security) should differ from workspace admins (platform). Use Key Vault RBAC and PIM for key operations.
6. **Audit key use**
   Send Key Vault diagnostic logs to Log Analytics and alert on disable, delete and unexpected unwrap failures.

04 Check your understanding

Five questions. The answer key follows the questions.

1 Which network hop in Azure Databricks is NOT encrypted by default?

- **A** Browser or API client to the control plane
- **B** Control plane to classic compute
- **C** Compute to ADLS Gen2 storage
- **D** Worker to worker (Spark shuffle)

2 What does switching to a customer-managed key change?

- **A** It turns on encryption at rest, which is off by default
- **B** Who owns the key and who can revoke it
- **C** The algorithm, from AES-128 to AES-256
- **D** It encrypts data in transit as well as at rest

3 Which CMK scope protects notebook source, SQL query history and Databricks-backed secrets?

- **A** Managed services
- **B** Workspace storage
- **C** Managed disks
- **D** The ADLS Gen2 storage account

4 With envelope encryption, what happens when you rotate the CMK?

- **A** All stored data is decrypted and re-encrypted immediately
- **B** The data-encryption key is re-wrapped; the data itself is not re-encrypted
- **C** Data encrypted with the old version is deleted
- **D** Key Vault is no longer needed after rotation

5 **The CMK used for workspace storage is deleted and purged from Key Vault. What happens?**

A Databricks falls back to a platform-managed key

B Microsoft support can restore the key on request

C The data encrypted with it becomes permanently unreadable

D Only data written after the deletion is affected

ANSWER KEY

1	**D**	Shuffle traffic between workers stays inside your VNet and is not encrypted by default. Every other hop uses TLS 1.2+. Enable inter-node encryption with an init script if policy requires it.
2	**B**	Data is already encrypted with AES-256 by default. A CMK moves the key into your Key Vault, so you control its use, rotation and revocation.
3	**A**	These objects live in the Databricks control plane, which is covered by the managed services key.
4	**B**	Only the small data-encryption key is wrapped by your key, so rotation re-wraps that key. This is why rotation is quick even for terabytes of data.
5	**C**	Purging the key is crypto-shredding: nobody can recover the data. This is why soft delete and purge protection are required on the vault.

Encryption is on by default

At rest and in transit, every zone is encrypted without configuration. CMK changes who holds the key, not whether data is encrypted.

CMK is an operational commitment

Choose it when a regulation or customer requires key control. Then protect, monitor and rotate the key like production infrastructure.